Multiplicative Quadrature via Logarithmic Means (MQLM): From Log-Linear Interpolation to High-Order Numerical Integration

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Abstract

This paper studies Multiplicative Quadrature via Logarithmic Means (MQLM), a quadrature rule for positive integrands based on the logarithmic mean. The rule is obtained by integrating exactly the piecewise log-linear interpolant of the nodal data, in direct analogy with the trapezoidal rule as the exact integral of the piecewise linear interpolant. Local and global exactness on log-affine functions are proved, explicit second- and fourth-order asymptotic coefficients are derived for the base rule when $f = e^u$, and it is shown that Richardson extrapolation produces a fourth-order rule with an explicit leading term. Numerical experiments indicate that the extrapolated MQLM rule is highly competitive with Simpson's rule on positive log-smooth functions, especially for exponential and near-exponential profiles. The method is specialized rather than universal, but it has a clear structural advantage when $\log f$ is substantially more regular than f itself.

1 Introduction

Classical composite quadrature rules are usually constructed from polynomial interpolation. The trapezoidal rule is the exact integral of a piecewise linear interpolant, while Simpson's rule arises from piecewise quadratic interpolation; this point of view is standard in numerical analysis [1, 3, 5].

For many positive integrands, however, the more natural object is not f itself but $\log f$. This is especially true for functions of the form

$$f(x) = e^{u(x)},$$

which occur frequently in analysis, probability, and mathematical physics. When $\log f$ is smoother, more nearly linear, or simply more informative than f, additive interpolation may fail to capture the relevant local structure.

The present work develops Multiplicative Quadrature via Logarithmic Means (MQLM), a quadrature rule adapted to this setting. The key idea is simple: interpolate $\log f$ linearly on each subinterval, exponentiate, and integrate the resulting log-linear interpolant exactly. The resulting formula is expressed in terms of the logarithmic mean, a classical two-point mean [6]. In this sense, the proposed MQLM rule is the multiplicative counterpart of the trapezoidal rule.

This places the method within the standard interpolation-based philosophy of quadrature [1, 2, 4], but with the interpolation performed in $\log f$ rather than in f. The viewpoint adopted here treats the MQLM rule as the exact integral of the piecewise log-linear interpolant and uses this construction to derive an explicit asymptotic comparison with the trapezoidal rule through the decomposition $f'' = f((u')^2 + u'')$. Even where related ideas overlap with broader traditions of structure-adapted approximation and extrapolation [7, 8], this formulation gives a concise and mathematically transparent framework for the present analysis.

The main contributions of the paper are:

- a construction of the MQLM rule as the exact integral of the piecewise log-linear interpolant;
- local and global exactness for log-affine functions;
- a second-order global asymptotic expansion

$$Q_L(h) = \int_a^b f(x)\,dx + \frac{h^2}{12}\int_a^b f(x)u''(x)\,dx + O(h^4), \qquad f = e^u,$$

  on uniform meshes;
- an explicit fourth-order coefficient for the base rule, revealing the next layer of log-geometric structure;
- a fourth-order extrapolated rule obtained by Richardson acceleration, together with its leading asymptotic term.

The method should be viewed as a specialized rule for positive functions with multiplicative structure, not as a universal replacement for classical formulas. Its strongest performance is expected when a log-linear local model is more faithful than an additive polynomial one.

2 Setting and Notation

Throughout the paper, $f : [a, b] \to (0, \infty)$ denotes a positive integrand and

$$I = \int_a^b f(x)\,dx$$

is the target integral.

For a general partition

$$a = x_0 < x_1 < \cdots < x_N = b,$$

write

$$h_i = x_{i+1} - x_i, \qquad f_i = f(x_i).$$

The logarithmic mean is used:

$$L(a, b) = \frac{b - a}{\log b - \log a}, \qquad a, b > 0,\ a \neq b,$$

with the continuous extension $L(a, a) = a$.

The corresponding composite MQLM rule on a general partition is

$$Q_L(\{x_i\}) = \sum_{i=0}^{N-1} h_i\, L(f_i, f_{i+1}).$$

For the asymptotic analysis, the discussion is restricted to a uniform mesh $x_i = a + ih$, where

$$h = \frac{b - a}{N},$$

and write

$$Q_L(h) = \sum_{i=0}^{N-1} h\, L(f_i, f_{i+1}).$$

Whenever $f = e^u$, regularity assumptions are imposed on u. In particular, the asymptotic result proved below assumes $u \in C^4([a, b])$.

3 Log-Linear Interpolation Viewpoint

The central interpretation of the method is interpolation in the variable $\log f$. Given nodal values $f_i > 0$, define the piecewise log-linear interpolant $\widetilde{f}$ by

$$\log \widetilde{f}(x) = \frac{x_{i+1} - x}{h_i} \log f_i + \frac{x - x_i}{h_i} \log f_{i+1}, \qquad x \in [x_i, x_{i+1}].$$

Equivalently,

$$\widetilde{f}(x) = f_i^{(x_{i+1}-x)/h_i} f_{i+1}^{(x-x_i)/h_i}.$$

This is the multiplicative analogue of the piecewise linear interpolant behind the trapezoidal rule.

Theorem 1 (Local Exactness for Log-Affine Functions)**.** *Let $f : [x_i, x_{i+1}] \to (0, \infty)$ be such that $\log f$ is affine on $[x_i, x_{i+1}]$. Equivalently,*

$$f(x) = e^{\alpha x + \beta}$$

for some $\alpha, \beta \in \mathbb{R}$. Then

$$h_i \, L(f(x_i), f(x_{i+1})) = \int_{x_i}^{x_{i+1}} f(x) \, dx.$$

Proof. Let $f_i = e^{\alpha x_i + \beta}$ and $f_{i+1} = e^{\alpha x_{i+1} + \beta}$. Then

$$h_i \, L(f_i, f_{i+1}) = h_i \, \frac{f_{i+1} - f_i}{\log f_{i+1} - \log f_i} = h_i \, \frac{f_{i+1} - f_i}{\alpha(x_{i+1} - x_i)} = \frac{f_{i+1} - f_i}{\alpha}.$$

On the other hand,

$$\int_{x_i}^{x_{i+1}} e^{\alpha x + \beta} \, dx = \frac{1}{\alpha} \Big(e^{\alpha x_{i+1} + \beta} - e^{\alpha x_i + \beta} \Big) = \frac{f_{i+1} - f_i}{\alpha}.$$

The case $\alpha = 0$ is immediate, since both sides equal $h_i e^{\beta}$. □

Corollary 1 (Global Exactness for Log-Affine Functions)**.** *If $\log f(x) = \alpha x + \beta$ on $[a, b]$, then for every partition*

$$Q_L(\{x_i\}) = \int_a^b f(x) \, dx.$$

In particular, $Q_L(h)$ is exact on every uniform mesh.

Proof. Apply Theorem 1 on each subinterval and sum. □

Proposition 1 (Exactness on the Piecewise Log-Linear Interpolant)**.** *Let $\widetilde{f}$ be the piecewise log-linear interpolant defined above. Then*

$$Q_L(\{x_i\}) = \int_a^b \widetilde{f}(x) \, dx.$$

Proof. On each subinterval $[x_i, x_{i+1}]$, the function $\widetilde{f}$ is log-affine. The claim therefore follows from Theorem 1 by summation. □

4 Geometric Motivation

The geometric rule

$$Q_G = \sum_{i=0}^{N-1} h_i \sqrt{f_i f_{i+1}}$$

appears naturally as a simpler log-space approximation. Indeed, define the piecewise constant function

$$\widetilde{f}_G(x) = \exp\left(\frac{\log f_i + \log f_{i+1}}{2}\right) = \sqrt{f_i f_{i+1}}, \qquad x \in [x_i, x_{i+1}].$$

Then

$$Q_G = \int_a^b \widetilde{f}_G(x)\, dx.$$

Thus Q_G is a zeroth-order log-space construction, whereas Q_L integrates the first-order log-linear interpolant exactly.

5 Asymptotic Analysis

The discussion now turns from construction to error analysis. The next theorem is an asymptotic statement on *uniform meshes*. Its proof is local-to-global: one first expands the rule and the exact integral on a single subinterval around the midpoint, then sums the local errors and identifies the resulting midpoint sum with an integral.

Theorem 2 (Second-Order Global Accuracy)**.** *Let $f(x) = e^{u(x)}$ with $u \in C^4([a,b])$ and $f(x) > 0$ on $[a,b]$. Then, on a uniform mesh,*

$$Q_L(h) = \int_a^b f(x)\, dx + c_2 h^2 + O(h^4),$$

where

$$c_2 = \frac{1}{12} \int_a^b f(x) u''(x)\, dx.$$

In particular, $Q_L(h)$ is second-order accurate.

Proof. Fix a subinterval $[x_i, x_{i+1}]$, let $m_i = (x_i + x_{i+1})/2$, and set

$$u_k = u^{(k)}(m_i), \qquad f(m_i) = e^{u_0}.$$

Because $u \in C^4([a,b])$, all $O(\cdot)$ terms below are uniform in i. Taylor expansion around m_i gives

$$u\left(m_i \pm \frac{h}{2}\right) = u_0 \pm \frac{u_1 h}{2} + \frac{u_2 h^2}{8} \pm \frac{u_3 h^3}{48} + \frac{u_4 h^4}{384} + O(h^5).$$

Write

$$a_i = u\left(m_i - \frac{h}{2}\right), \qquad b_i = u\left(m_i + \frac{h}{2}\right).$$

Then

$$\frac{a_i + b_i}{2} = u_0 + \frac{u_2 h^2}{8} + O(h^4), \qquad b_i - a_i = u_1 h + \frac{u_3 h^3}{24} + O(h^5).$$

The logarithmic contribution on this subinterval is

$$Q_{L,i} = h\,\frac{e^{b_i} - e^{a_i}}{b_i - a_i} = h\,e^{(a_i+b_i)/2}\,\frac{2\sinh((b_i-a_i)/2)}{b_i - a_i}.$$

Since

$$e^{(a_i+b_i)/2} = e^{u_0}\left(1 + \frac{u_2 h^2}{8} + O(h^4)\right)$$

and

$$\frac{2\sinh(z/2)}{z} = 1 + \frac{z^2}{24} + O(z^4), \qquad z \to 0,$$

with $z = b_i - a_i = u_1 h + O(h^3)$, it follows that

$$Q_{L,i} = h\,e^{u_0}\left(1 + \frac{(3u_2 + u_1^2)h^2}{24} + O(h^4)\right).$$

For the exact integral, writing $x = m_i + s$ with $-h/2 \le s \le h/2$,

$$u(m_i + s) = u_0 + u_1 s + \frac{u_2}{2}s^2 + O(h^3),$$

hence

$$e^{u(m_i+s)} = e^{u_0}\left(1 + u_1 s + \frac{u_2 + u_1^2}{2}s^2 + O(h^3)\right).$$

Integrating, the odd term vanishes and

$$\int_{x_i}^{x_{i+1}} f(x)\,dx = h\,e^{u_0}\left(1 + \frac{(u_2 + u_1^2)h^2}{24} + O(h^4)\right).$$

Therefore

$$Q_{L,i} - \int_{x_i}^{x_{i+1}} f(x)\,dx = \frac{h^3}{12} f(m_i)u''(m_i) + O(h^5).$$

Summing over i yields

$$Q_L(h) - \int_a^b f(x)\,dx = \frac{h^3}{12}\sum_{i=0}^{N-1} f(m_i)u''(m_i) + O(Nh^5).$$

Since $N = (b-a)/h$, the remainder is $O(h^4)$. Finally,

$$h\sum_{i=0}^{N-1} f(m_i)u''(m_i) = \int_a^b f(x)u''(x)\,dx + O(h^2)$$

by the midpoint rule applied to the smooth function $f\,u''$. Substituting gives the result. □

Theorem 3 (Fourth-Order Coefficient of the Base Rule)**.** *Let $f(x) = e^{u(x)}$ with $u \in C^6([a,b])$ and $f(x) > 0$ on $[a,b]$. Then, on a uniform mesh,*

$$Q_L(h) = \int_a^b f(x)\,dx + c_2 h^2 + c_4 h^4 + O(h^6),$$

where

$$c_2 = \frac{1}{12}\int_a^b f(x)u''(x)\,dx$$

and

$$c_4 = \int_a^b f(x)\left(\frac{u''''(x)}{480} + \frac{(u''(x))^2}{160} + \frac{u'(x)u'''(x)}{720} + \frac{(u'(x))^2u''(x)}{480}\right)dx$$
$$- \frac{1}{288}\Big[f(x)\big(u'(x)u''(x) + u'''(x)\big)\Big]_a^b.$$

Proof. Fix a subinterval $[x_i, x_{i+1}]$, let $m_i = (x_i + x_{i+1})/2$, and write

$$u_k = u^{(k)}(m_i), \qquad f(m_i) = e^{u_0}.$$

Because $u \in C^6([a,b])$, all remainder terms below are uniform in i. Expanding around m_i,

$$u\left(m_i \pm \frac{h}{2}\right) = u_0 \pm \frac{u_1 h}{2} + \frac{u_2 h^2}{8} \pm \frac{u_3 h^3}{48} + \frac{u_4 h^4}{384} \pm \frac{u_5 h^5}{3840} + O(h^6).$$

Set

$$a_i = u\left(m_i - \frac{h}{2}\right), \qquad b_i = u\left(m_i + \frac{h}{2}\right).$$

Then

$$\frac{a_i + b_i}{2} = u_0 + \frac{u_2 h^2}{8} + \frac{u_4 h^4}{384} + O(h^6),$$
$$b_i - a_i = u_1 h + \frac{u_3 h^3}{24} + \frac{u_5 h^5}{1920} + O(h^7).$$

The logarithmic contribution is

$$Q_{L,i} = h\,\frac{e^{b_i} - e^{a_i}}{b_i - a_i} = h\,e^{(a_i+b_i)/2}\,\frac{2\sinh((b_i - a_i)/2)}{b_i - a_i}.$$

Now

$$e^{(a_i+b_i)/2} = e^{u_0}\left(1 + \frac{u_2 h^2}{8} + \left(\frac{u_4}{384} + \frac{u_2^2}{128}\right)h^4 + O(h^6)\right),$$

while

$$\frac{2\sinh(z/2)}{z} = 1 + \frac{z^2}{24} + \frac{z^4}{1920} + O(z^6).$$

With $z = b_i - a_i = u_1 h + \frac{u_3 h^3}{24} + O(h^5)$, this gives

$$\frac{2\sinh((b_i - a_i)/2)}{b_i - a_i} = 1 + \frac{u_1^2 h^2}{24} + \left(\frac{u_1 u_3}{288} + \frac{u_1^4}{1920}\right)h^4 + O(h^6).$$

Multiplying the two series, one obtains

$$Q_{L,i} = h\,e^{u_0}\left(1 + \frac{(3u_2 + u_1^2)h^2}{24} + \left(\frac{u_4}{384} + \frac{u_2^2}{128} + \frac{u_1 u_3}{288} + \frac{u_1^2 u_2}{192} + \frac{u_1^4}{1920}\right)h^4 + O(h^6)\right).$$

For the exact integral, write $x = m_i + s$, $-h/2 \le s \le h/2$. Then

$$u(m_i + s) = u_0 + u_1 s + \frac{u_2}{2}s^2 + \frac{u_3}{6}s^3 + \frac{u_4}{24}s^4 + O(h^5),$$

hence

$$e^{u(m_i+s)} = e^{u_0}\left(1 + u_1 s + \frac{u_2+u_1^2}{2}s^2 + \left(\frac{u_4}{24} + \frac{u_2^2}{8} + \frac{u_1u_3}{6} + \frac{u_1^2u_2}{4} + \frac{u_1^4}{24}\right)s^4 + O(h^5)\right).$$

After integration over the symmetric interval, the odd terms vanish and

$$\int_{x_i}^{x_{i+1}} f(x)\,dx = h\,e^{u_0}\left(1 + \frac{(u_2+u_1^2)h^2}{24} + \left(\frac{u_4}{1920} + \frac{u_2^2}{640} + \frac{u_1u_3}{480} + \frac{u_1^2u_2}{320} + \frac{u_1^4}{1920}\right)h^4 + O(h^6)\right),$$

since

$$\int_{-h/2}^{h/2} s^2\,ds = \frac{h^3}{12}, \qquad \int_{-h/2}^{h/2} s^4\,ds = \frac{h^5}{80}.$$

Subtracting the two expansions yields

$$Q_{L,i} - \int_{x_i}^{x_{i+1}} f(x)\,dx = \frac{h^3}{12}f(m_i)u''(m_i) + h^5 B(m_i) + O(h^7),$$

where

$$B(x) = f(x)\left(\frac{u''''(x)}{480} + \frac{(u''(x))^2}{160} + \frac{u'(x)u'''(x)}{720} + \frac{(u'(x))^2u''(x)}{480}\right).$$

Equivalently, with

$$A(x) = \frac{1}{12}f(x)u''(x),$$

the local error reads

$$Q_{L,i} - \int_{x_i}^{x_{i+1}} f(x)\,dx = h^3 A(m_i) + h^5 B(m_i) + O(h^7).$$

Summing over the mesh yields

$$Q_L(h) - \int_a^b f(x)\,dx = h^2\left(h\sum_{i=0}^{N-1} A(m_i)\right) + h^4\left(h\sum_{i=0}^{N-1} B(m_i)\right) + O(h^6).$$

Applying the midpoint expansion

$$h\sum_{i=0}^{N-1} A(m_i) = \int_a^b A(x)\,dx - \frac{h^2}{24}\big(A'(b) - A'(a)\big) + O(h^4)$$

and

$$h\sum_{i=0}^{N-1} B(m_i) = \int_a^b B(x)\,dx + O(h^2),$$

one obtains

$$Q_L(h) - \int_a^b f(x)\,dx = h^2\int_a^b A(x)\,dx + h^4\left(\int_a^b B(x)\,dx - \frac{A'(b)-A'(a)}{24}\right) + O(h^6).$$

Since

$$A'(x) = \frac{1}{12}(fu'')'(x) = \frac{1}{12}f(x)\big(u'(x)u''(x) + u'''(x)\big),$$

the stated formula for c_4 follows. □

Proposition 2 (Structural Comparison with the Trapezoidal Rule)**.** *Under the hypotheses of Theorem 2,*

$$Q_T(h) = \int_a^b f(x)\,dx + \frac{h^2}{12}\int_a^b f(x)\big((u'(x))^2 + u''(x)\big)\,dx + O(h^4),$$

whereas

$$Q_L(h) = \int_a^b f(x)\,dx + \frac{h^2}{12}\int_a^b f(x)u''(x)\,dx + O(h^4).$$

Thus the MQLM rule removes the $(u')^2$ contribution from the dominant trapezoidal error term.

Proof. The Euler–Maclaurin expansion gives

$$Q_T(h) - \int_a^b f(x)\,dx = \frac{h^2}{12}\int_a^b f''(x)\,dx + O(h^4).$$

Since $f = e^u$,

$$f''(x) = f(x)\big((u'(x))^2 + u''(x)\big).$$

Substitute this identity and compare with Theorem 2. □

Corollary 2 (Fourth-Order Richardson Extrapolation)**.** *Under the hypotheses of Theorem 3, define*

$$Q_L^{(4)}(h) = \frac{4Q_L(h/2) - Q_L(h)}{3}.$$

Then

$$Q_L^{(4)}(h) = \int_a^b f(x)\,dx - \frac{c_4}{4}h^4 + O(h^6).$$

Proof. By Theorem 3,

$$Q_L(h) = I + c_2h^2 + c_4h^4 + O(h^6), \qquad Q_L(h/2) = I + c_2\frac{h^2}{4} + c_4\frac{h^4}{16} + O(h^6).$$

Hence

$$4Q_L(h/2) - Q_L(h) = 3I - \frac{3}{4}c_4h^4 + O(h^6),$$

and division by 3 gives the claim. □

6 Asymptotic Interpretation

Theorem 2 shows that the leading error of the MQLM rule is governed by the curvature of $\log f$:

$$Q_L(h) - I = \frac{h^2}{12}\int_a^b f(x)u''(x)\,dx + O(h^4), \qquad f = e^u.$$

By Proposition 2, the trapezoidal rule depends instead on

$$f'' = f\big((u')^2 + u''\big),$$

so its dominant error contains both the slope term $(u')^2$ and the curvature term u''. The MQLM rule removes the $(u')^2$ term entirely. This is the precise structural reason for its strong behavior on exponential and near-exponential profiles.

Theorem 3 shows that the next coefficient already mixes genuinely nonlinear log-geometry: besides u'''', it contains the quadratic terms $(u'')^2$ and $(u')^2u''$, as well as a boundary correction. Thus the MQLM rule is not merely a first-order correction to trapezoidal behavior; it carries a higher-order structure naturally expressed in derivatives of $u = \log f$.

Corollary 2 shows that this higher-order structure is inherited directly by the extrapolated rule: Richardson acceleration removes the h^2 term, but its remaining h^4 behavior is still governed by the explicit coefficient c_4 of the base MQLM rule.

7 Numerical Experiments

The experiments were designed to test the theoretical claims above:

- exactness on log-affine profiles;
- second-order behavior of the base MQLM rule;
- fourth-order behavior after Richardson extrapolation.

The benchmark included 45 test functions in 9 families, with uniform grids

$$N \in \{8, 16, 32, 64, 128, 256, 512, 1024\}.$$

Unless explicitly indicated in the function label, the integration interval was $[0, 1]$. High-precision reference values were computed with arbitrary-precision integration using `mpmath`. Each reference integral was recomputed at increasing decimal precision until two successive runs agreed on the digits reported in the benchmark tables. The methods compared were Midpoint, Trapezoidal, Simpson, Geometric, Geometric + Richardson, MQLM, MQLM + Richardson, Gauss–Legendre, and Romberg.

The test suite included pure exponentials, exponential compositions, positive rational profiles, steep positive profiles, and adverse cases such as endpoint singularities or oscillatory behavior. The numerical section is therefore not just a leaderboard: it is an empirical check of the structural predictions of the theory.

7.1 Benchmark Design

The benchmark was organized by family in order to test different structural regimes of the theory:

- **exponential**: exactly or nearly log-affine profiles;
- **exp_composite**: positive functions of the form $e^{\phi(x)}$, where the log-linear viewpoint is natural but not exact;
- **poly_positive**, **algebraic_positive**, and **classic_positive**: smooth positive functions not primarily designed from an exponential ansatz;
- **rational_positive** and **steep_positive**: positive profiles with sharp variation or narrow peaks;
- **osc_positive**: positive but oscillatory functions, where the theory predicts no universal advantage;

- **endpoint_positive**: endpoint-singular functions, included precisely because they violate the smoothness assumptions behind the asymptotic results.

For the composite methods, observed orders were estimated from successive mesh refinements by

$$p_{\text{obs}}(N) = \log_2\left(\frac{E_N}{E_{2N}}\right),$$

where E_N denotes the absolute error on the uniform mesh with N subintervals. These orders were reported only when both successive errors were finite and nonzero. For the global comparators, the benchmark used Gauss–Legendre rules with 32 and 64 nodes and a 7-level Romberg scheme; for these methods the tables report absolute error and evaluation counts rather than mesh-based observed order. All composite methods were run on the same uniform meshes listed above, so the comparison among Midpoint, Trapezoidal, Simpson, Geometric, and MQLM variants is pointwise in N and uses the same nodal sampling pattern. This matters because the main theoretical claims concern uniform-mesh asymptotics.

7.2 Benchmark Catalogue

Table 1 lists the families and functions used in the heavy benchmark underlying the summary statistics reported below.

Family	Functions
algebraic_positive	$(1+x)^{1/2}$, $(1+x)^{3/2}$, $\cosh(x)$, $\cosh(3x)$
classic_positive	$1+0.5x+0.1x^2+e^x$, $4/(1+x^2)$, e^{-x^2} on $[-2,2]$
endpoint_positive	$\sqrt{x}$, $\sqrt{x}e^x$, $x^{1/3}$, $x^{1/3}(1+x)$, $x^{3/2}$
exp_composite	e^{-20x^2}, e^{-x^2}, e^{5x^2}, $e^{\cos x}$, $e^{\sin 3x}$, $e^{\sin x}$, e^{x^2}, e^{x+x^2}
exponential	10^x, e^{-5x}, e^{20x}, e^{2x}, e^x
osc_positive	$1+\sin x$, $2+\cos(10x)$, $2+\cos(20x)$, $e^{\sin 20x}$
poly_positive	$(1+x)^5$, $1+3x-2x^2+x^4$, $1+x$, $1+x^2$, $1+x^3$, $1+x^4$
rational_positive	$1/(1+100x^2)$, $1/(1+25x^2)$, $1/(1+x)$, $1/\sqrt{1+x}$
steep_positive	$1/(0.01+(x-0.4)^2)$, $1/\sqrt{0.01+(x-0.5)^2}$, $e^{-100(x-0.5)^2}$, $e^{-400(x-0.2)^2}$, e^{100x}, e^{50x}

Table 1: Function catalogue used in the heavy benchmark. Unless explicitly indicated, the integration interval is $[0,1]$.

7.3 Model Examples

Three model examples are particularly informative.

- For $f(x) = e^x$, Theorem 1 predicts exactness on every subinterval. In the benchmark, the base MQLM rule at $N = 32, 64, 128, 256$ produced absolute error 2.22×10^{-16} throughout, which is consistent with exactness up to floating-point roundoff.

- For $f(x) = e^{x^2}$, one has $u''(x) = 2$, so Theorem 2 predicts a positive leading error term. Numerically, the base MQLM rule overestimates the integral with errors 2.38×10^{-4}, 5.95×10^{-5}, and 1.49×10^{-5} at $N = 32, 64, 128$, exhibiting both the predicted sign and a clear second-order reduction factor.

- For $f(x) = e^{-x^2}$, one has $u''(x) = -2$, so Theorem 2 predicts a negative leading error term. Numerically, the base MQLM rule underestimates the integral with errors 1.22×10^{-4}, 3.04×10^{-5}, and 7.60×10^{-6} at $N = 32, 64, 128$, again with the predicted sign and second-order scaling.

For the Richardson-extrapolated rule, the same two Gaussian-type examples display fourth-order behavior: for e^{x^2} the absolute error drops from 2.21×10^{-9} at $N = 32$ to 1.38×10^{-10} at $N = 64$, while for e^{-x^2} it drops from 2.48×10^{-9} to 1.55×10^{-10}. These examples test not only order, but also the sign and structure of the asymptotic mechanism identified in Theorem 2.

The strengthened corollary can also be checked directly. Evaluating the coefficient c_4 from Theorem 3 for these two examples gives

$$c_4\big(e^{x^2}\big) \approx 9.275963 \times 10^{-3}, \qquad c_4\big(e^{-x^2}\big) \approx 1.040329 \times 10^{-2}.$$

Therefore Corollary 2 predicts

$$Q_L^{(4)}(h) - I \approx -\frac{c_4}{4} h^4,$$

so at $N = 32$ and $N = 64$ one expects, respectively,

$$e^{x^2}: \quad -2.21156 \times 10^{-9},\ -1.38223 \times 10^{-10},$$

$$e^{-x^2}: \quad -2.48034 \times 10^{-9},\ -1.55021 \times 10^{-10}.$$

The observed signed errors are

$$e^{x^2}: \quad -2.21249 \times 10^{-9},\ -1.38237 \times 10^{-10},$$

$$e^{-x^2}: \quad -2.48014 \times 10^{-9},\ -1.55018 \times 10^{-10},$$

The agreement is close in both sign and magnitude, so the numerical behavior of the extrapolated rule is consistent not only with fourth-order convergence, but with the explicit leading term predicted by the theory.

8 Global Results

At the finest grid, the median errors for the main uniform-grid methods were:

Method	Median Error	Order
MQLM + Richardson	1.82×10^{-14}	≈ 4
Geometric + Richardson	4.67×10^{-14}	≈ 4
Simpson	2.75×10^{-13}	≈ 4

The extrapolated MQLM rule achieved the smallest median error among the uniform-grid methods in the benchmark. This is consistent with Corollary 2: once the base rule captures the correct log-linear local behavior, Richardson extrapolation removes the leading h^2 term and leaves an h^4 regime governed by the explicit coefficient inherited from the base expansion.

9 Comparison with Simpson

On valid positive test cases, MQLM + Richardson outperformed Simpson in 23 of 38 cases, with median improvement factor approximately 5.4. This indicates a frequent advantage on a relevant subclass of positive log-smooth integrands, not a universal superiority claim. The comparison is consistent with the interpolation viewpoint: the MQLM rule is favored when a log-linear local model is substantially more faithful than an additive polynomial one.

10 Where the Method Excels

The strongest performance was observed for exponential functions, exponential compositions, several positive rational profiles, and positive log-smooth functions more broadly. These are precisely the situations in which the logarithm of the integrand is often more regular than the integrand itself, and where the leading error is better interpreted through u'' than through f''.

11 Limitations

The method is not universal. It performs worse in settings such as sign-changing functions, endpoint singularities, and extreme-curvature cases where extrapolation becomes unstable. From the theoretical viewpoint, these are exactly the situations where the log-linear model breaks down or where the smoothness assumptions behind Theorem 2 cease to be informative.

12 Discussion

The results support the following picture:

- classical quadrature rules are naturally tied to additive interpolation;
- the MQLM rule is naturally tied to multiplicative, or log-linear, interpolation;
- for positive exponential-type data, the latter can be structurally more appropriate.

This places the method within the general tradition of structure-adapted numerical approximation and extrapolation rather than in competition with classical formulas in every regime.

13 Conclusion

This paper introduces Multiplicative Quadrature via Logarithmic Means (MQLM) and shows that it is best understood as the exact integral of the piecewise log-linear interpolant of nodal data. The MQLM rule is exact on log-affine functions, has an explicit second-order asymptotic coefficient governed by u'' when $f = e^u$, and admits an explicit fourth-order coefficient for the base rule that reveals the next layer of log-geometric structure. After Richardson extrapolation, the method reaches fourth order with a leading term inherited directly from that same coefficient, and the benchmark confirms this behavior on representative Gaussian-type examples.

The method has clear mathematical interest and practical promise on positive integrands with multiplicative structure. Its natural scope is specialized rather than universal: it is most compelling when a log-linear local model is superior to an additive polynomial one.

Future work includes adaptive variants, hybrid schemes, stronger comparison results against classical rules, and possible multidimensional extensions.

Declarations

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Data availability Benchmark data are available from the author upon request.

Code availability Implementation and benchmark scripts are made available by the author under the GNU AGPLv3 license in the project repository.

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